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Review article

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Human organoids as 3D in vitro platforms for drug discovery: opportunities and challenges

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Abstract

Organoids are 3D structures derived from stem cells that recapitulate key architectural and functional aspects of the corresponding tissue. Compared with conventional 2D cell lines, human organoids provide experimental models that more closely reflect human physiology. Their ability to capture the complexity and heterogeneity of human tissues enables the study of disease mechanisms, drug efficacy and toxicity. When generated from patient material, organoids also allow the assessment of individual drug responses. In this Review, we explore the utility of organoids in drug discovery. We outline current methodologies for generating and maintaining organoids, examine their applications in disease modelling, drug screening and safety evaluation, and consider regulatory aspects and the challenges for their broader adoption in drug discovery.

Sections

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Introduction

In drug discovery, in vitro models have a pivotal role by providing defined experimental systems for the investigation of cellular responses to pharmacological agents. Human cell lines have traditionally served as drug discovery workhorses, owing to their relative simplicity, low cost and broad accessibility across research laboratories. Cell lines are typically derived from tumours and have been adapted to grow on 2D plastic surfaces. As a consequence, cell lines do not represent tissue architecture, nor do they contain the complement of functional cell types of individual organs. This has led to the exploration of more physiological 3D culture systems, such as organoids, that more precisely mimic the in vivo microenvironment. Compared with animal experimentation, organoids are more cost-effective and enable high-throughput screening (HTS), while avoiding the species differences and ethical concerns associated with animal testing. Mina Bissell and colleagues had a major role in pioneering the 3D culture of epithelia through the use of Matrigel, an extracellular matrix (ECM)-like hydrogel. In the context of 3D Matrigel culture, cells engage in more physiological cell-to-cell and cell-to-matrix interactions. Indeed, Bissell could show that breast-derived epithelial cells form 3D glandular structures in Matrigel. These organoids could even be coerced to secrete milk[1].

The development of organoid technology using pluripotent stem cells (PS cells) has its roots in the discovery of embryonic stem (ES) cells in the 1980s[2,3]. ES cells exist naturally only in blastula-stage embryos yet can be cultured long term under defined conditions and, in theory, can be instructed to differentiate into any cell type — such as neurons[4] or myocardiocytes[5] — under the right growth factor conditions. Logistical and ethical challenges have complicated the widespread use of human ES cells. These challenges are directly related to the fact that ES cells derive from early human embryos and that this derivation leads to the destruction of the embryo. These issues were largely solved by the invention of induced pluripotent stem cells (iPS cells) in 2006 by Shinya Yamanaka, who reprogrammed somatic cells from adult tissues into iPS cells by transfection of just four transcription factors[6]. Research on PS cells has primarily used 2D culture approaches[7]. When scientists began to explore 3D culture techniques, such as spherical iPS cell aggregation, it was discovered that PS cells can spontaneously create more complex structures. The first PS cell-based organoid models emerged from Yoshiki Sasai's team, who created structures that resembled retinas and various other parts of the central nervous system (CNS)[8–11]. These structures self-organized to recapitulate key features of the actual organ, including cell-type diversity and spatial organization. Standardized protocols for the generation of PS cell-based organoids that represent various human tissues are being developed at ever-increasing rates. These protocols, typically running over weeks or months, start from small clumps of iPS cells and focus on optimizing nutrient supply while mimicking the sequential developmental signalling environments to shepherd the PS cells towards the tissue of interest (Fig. 1a). Popular PS cell-derived organoids can mimic brain[12], retina[13], lung[14–17], kidney[18], liver[19] and intestine[20] tissue.

Virtually every organ has the capacity to replace damaged or worn-out cells through the action of its tissue stem cells (TSCs), with the likely exception of the heart muscle, the kidney glomerulus and most parts of the CNS. Haematopoietic stem cells in bone marrow were the first TSCs to be discovered[21]. Later, epithelial TSCs were identified in tissues such as the skin[22], cornea[23] and the intestinal lining[24]. In 2009, we demonstrated that a single TSC derived from the intestinal epithelium can create ever-expanding epithelial organoids containing all relevant intestinal cell types, showcasing the inherent self-organizing capabilities of TSCs and their cellular offspring[25]. Similar to iPS cells, specific growth factors and ECM components are essential to guide TSC expansion and differentiation into the various cell types of interest (Fig. 1b). As a main difference from the PS cell-based approach, TSC-based organoids are established from biopsy samples or other primary tissue sources (instead of from iPS cell-based cultures) and do not require in vitro specification to the tissue of interest.

The number of organoid studies has grown rapidly over the past decade, and the technology seems ready to start fulfilling its various promises for drug development[26]. Human organoids provide in vitro experimental models that can complement both traditional 2D human cell lines and animal models (Table 1). As stated above, cell lines poorly represent their tissue of origin: they have almost invariably undergone some form of malignant transformation, either because they derive from cancers or because they have extensively been adapted to 2D growth on plastic surfaces. Indeed, when transplanted, cell lines typically form tumours, whereas transplanted organoids produce healthy, functional tissue[27]. Compared with cell lines, animal models offer the advantage of recapitulating biological phenomena in the context of an entire organism, but they too have limitations: many human diseases remain difficult to reproduce in animals owing to inherent interspecies differences. Likewise, physiological differences between animals and humans limit the accurate prediction of drug safety and efficacy. Moreover, the use of animals is increasingly regulated, largely because of ethical considerations[28,29]. In 2022, the US Congress passed the FDA Modernization Act 2.0 (ref. 30), followed by Act 3.0 (ref. 31), to clarify the perceived requirement of animal models in drug development. More recently, the FDA has announced the intention to phase out animal testing in the development of monoclonal antibody therapies and other drugs[32]. Consequentially, regulatory authorities are actively exploring alternative approaches[28,29].

In this Review, we outline methodologies for the generation and maintenance of organoids and highlight their applications in disease modelling, therapeutic screening and the evaluation of drug pharmacology and toxicity. We also discuss their potential in personalized medicine, in which organoids can be derived from a patient's own cells, enabling individualized drug testing. Finally, we examine regulatory considerations and the challenges that must be addressed for organoid platforms to become established tools in drug development.

Disease modelling

Organoids have been applied across diverse disease areas, including genetic disorders, cancer and infectious diseases (Fig. 2).

Genetic disorders

Human organoids seem to be particularly well suited to model genetic disease. They can be derived either directly from biopsy samples of the affected organs of patients (patient-derived organoids, PDOs) or alternatively through CRISPR-mediated gene editing of organoids obtained from healthy individuals.

Cystic fibrosis (CF) is an autosomal recessive disorder caused by loss-of-function mutations in the *CFTR* gene[33,34]. This gene encodes the chloride/bicarbonate channel cystic fibrosis transmembrane conductance regulator (CFTR), which is crucial for ion and fluid homeostasis in multiple epithelial tissues, including lung and intestine[35]. Several CFTR modulators have been developed[36], for example, Orkambi, which consists of two compounds (ivacaftor and lumacaftor) and was developed

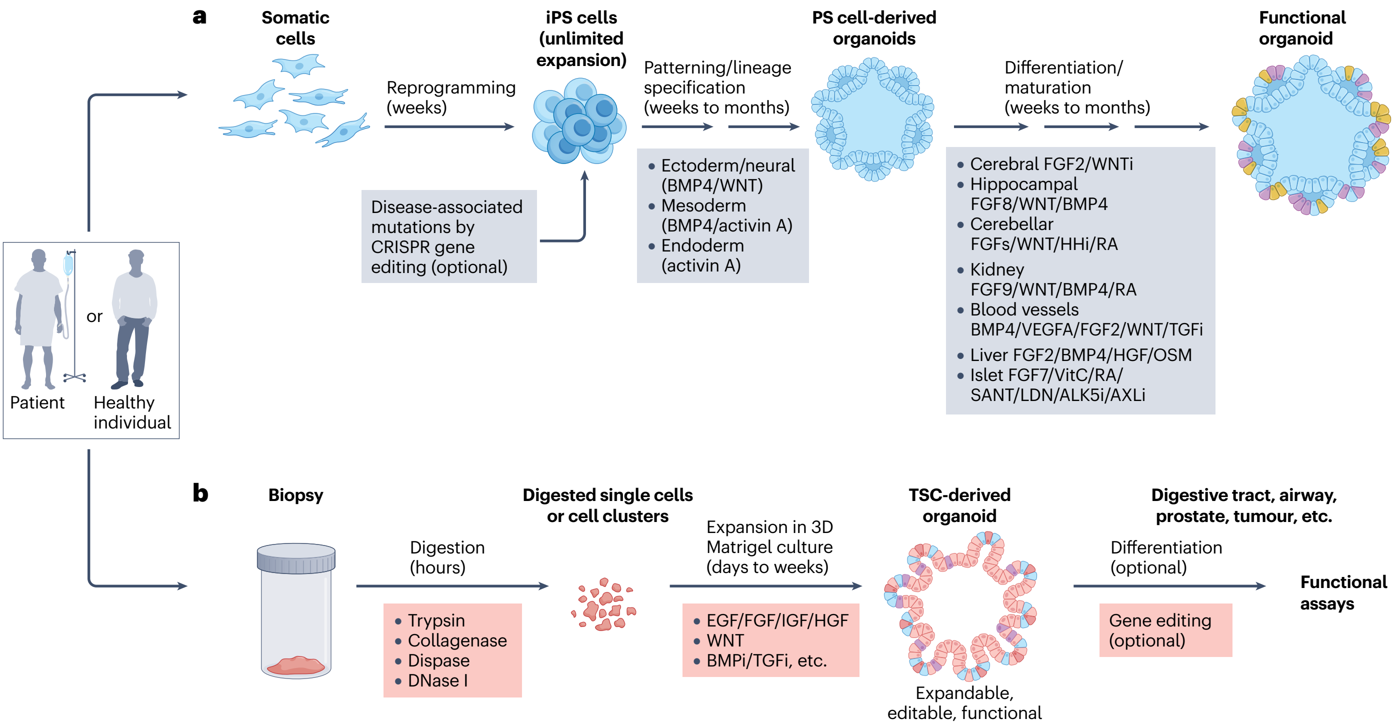

Fig. 1 | Pluripotent stem cell- and tissue stem cell-derived human organoids. **a**, Work-flow to generate inducible pluripotent stem cell (iPS cell)-derived organoids. Generation of organoids from iPS cells requires stepwise lineage specification that mimics in vivo embryonic development, progressing from pluripotency to the organ of interest. iPS cells are typically harvested as cell aggregates, which maintain cell–cell contact and spontaneously form complex structures. Organoids are first directed into a specific germ layer by applying appropriate niche factors, then further differentiated into organ-specific cell types using tissue-specific developmental cues. The resulting organoids commonly reach late fetal stages of development. Representative signalling factors that are involved in germ layer specification, as well as in cell-type-specific differentiation or maturation, are indicated. **b**, Work-flow to generate tissue stem cell (TSC)-derived organoids. Tissues are processed into single cells or small fragments, which are directly embedded in an extracellular matrix (such as Matrigel) for 3D culture. Because cells are derived from biopsy samples of the target organ, this bypasses the lineage specification steps essential for PS cell-derived organoids. TSC-based organoid cultures require only factors that support long-term expansion and as a result, they typically require less time to establish. Differentiation of functional cell types in TSC-derived organoids can occur spontaneously along with stem cell self-renewal, or can be further induced using cell-type-specific niche factors. TSC-derived organoids can be efficiently established from tissues that contain active stem cells, such as the digestive tract, whereas non-self-renewing tissues (brain, retina, heart muscle) can be generated only from iPS cells. Disease-associated genetic mutations can be introduced into the organoid model using CRISPR-based gene editing. This process requires edited cells to be highly proliferative to support their growth from a single cell, thus gene editing is typically carried out in iPS cells or TSCs, before lineage specification and cell differentiation. ALK, anaplastic lymphoma kinase; BMP, bone morphogenetic protein; EGF, epidermal growth factor; FGF, fibroblast growth factor; HGF, hepatocyte growth factor; HHi, hedgehog inhibitor; IGF, insulin-like growth factor; OSM, oncostatin M; RA, retinoic acid; TGF, transforming growth factor; VEGF, vascular endothelial growth factor; VitC, vitamin C.

specifically for patients with homozygous *CFTR*$^{\Delta508}$ mutations, encompassing an estimated 50% of the CF population[37,38]. Patient responses to this drug vary, even when carrying the same genetic mutation, as the activity of CFTR modulators is influenced by cellular and genetic backgrounds[39,40]. Notably, the genetic basis and resultant CF symptoms also vary considerably between humans and available mouse models. *Cftr*-mutant (including *Cftr*$^{\Delta508}$) mice do not develop the severe airway disease observed in patients with CF, such as mucus accumulation, airway obstruction or chronic lung infections[41]. CF was the first disease in which PDOs enabled a personalized medicine approach.

The CFTR channel is activated by cAMP. In healthy gut[42] or airway[43] organoids, addition of forskolin — a compound that raises cAMP by stimulating adenylate cyclase to convert ATP into cAMP — triggers rapid swelling as a result of CFTR-mediated water influx into the lumen. By contrast, organoids derived from patients with CF fail to swell in response to forskolin. This swelling failure can be rescued either by CRISPR-based correction of the *CFTR* mutations[44] or by treatment with effective CFTR modulators[45,46]. Importantly, drug responses measured in this swelling assay correlate well with clinical outcomes in patients, such as those measured by percentage of predicted forced expiratory volume in 1 s (ppFEV1) and sputum cell count (SCC)[47]. The simplicity of the organoid swelling readout also makes the assay suitable for drug HTS.

Primary ciliary dyskinesia (PCD), another genetic disorder that affects the airways, is caused by mutations in genes associated with cilium structure, including *DNAH5/11*, *DNAI1* and *CCDC39/40*, and can be explored using nasal airway organoids from patients with primary ciliary dyskinesia[48]. These organoids allow direct visualization of beating cilia and of structural changes by electron microscope imaging. The potential for gene therapy has been demonstrated in this organoid model, upon

Table 1 | Comparisons of human organoids with traditional preclinical models

Parameter	Human organoids	2D cell lines	Animal models
Physiological relevance	High: mimics human tissue complexity	Low: limited structural mimicry	High: whole organism context, with notable interspecies differences
Throughput	Medium/high: limited by growth complexity	High: easy to culture and analyse	Low: time-consuming studies
Cost/time	Medium: variable set-up and culture	Low: generally inexpensive	High: expensive and time-intensive
Scalability	Medium/high: technical challenges exist	High: easily scalable	Low: limited by animal availability
Regulatory acceptance	Emerging and increasingly recognized	Limited relevance	Established and widely accepted

correction of mutations in the *DNAH11* gene via prime editing[48]. Surfactant protein deficiency, caused by mutations in genes such as *SFTPB* (encodes pulmonary surfactant-associated protein B) and *SFTPC*, leads to familial pulmonary fibrosis. Surfactant proteins produced by alveolar type 2 (AT2) cells reduce surface tension at the air–liquid interface in the alveoli and thereby prevent alveolar collapse (atelectasis) during respiration, facilitating easier lung expansion and efficient gas exchange. Recently, AT2 cell organoids derived from fetal lung were reported[49], and iPS cell-derived lung organoids from a patient with *SFTPB* deficiency recapitulated the characteristic abnormal lamellar-body formation, which was rescued by re-expressing wild-type *SFTPB*[50].

Kidney diseases can also be modelled in organoids. Polycystic kidney disease (PKD), the most common inheritable renal disorder, has been studied using kidney organoids from patients with mutations in *PKD1* or *PKD2*, which encode polycystin 1 and polycystin 2, respectively. These organoids reproduce the hallmark feature of autosomal dominant PKD (ADPKD) – cyst formation – while also expressing the ADPKD gene signature[51,52]. HTS using scalable ADPKD organoids has identified promising compounds that reduce cyst growth, including quinazoline, a potential nuclear factor-κB (NF-κB) pathway modulator[51]. More recently, an organoid-on-a-chip platform (Box 1 and Fig. 3a) was established using iPS cell-derived kidney organoids carrying mutations in *PKHD1*, which encodes fibrocystin. This system mimics the fluid flow of the native tissue microenvironment and recapitulates cystogenesis in patients with autosomal recessive PKD (ARPKD), providing clinically relevant phenotypes for drug testing[53].

Genetic liver disorders. α1-Antitrypsin deficiency (AATD) is a monogenic disorder caused by mutations in the *SERPINA1* gene, resulting in low levels of α1-antitrypsin (A1AT)[54]. A1AT is primarily produced in the liver and serves as a serine protease inhibitor, protecting lung tissues from neutrophil elastase-mediated destruction. Patients with AATD have increased susceptibility to lung and liver damage[54,55], and liver organoids derived from patients with AATD model disease characteristics, including A1AT aggregation, diminished protein secretion and impaired elastase inhibition[56,57]. This model provides an opportunity to explore gene therapy and other strategies aimed at preventing A1AT polymerization and aggregation within hepatocytes.

Organoids generated from patients with *JAG1*-mutant Alagille syndrome show delayed differentiation into ductal cells and increased apoptosis in the lumen, reflecting in vivo abnormalities of the biliary tree[56]. Cholangiocyte organoids from patients with Wilson's disease, caused by mutations in the *ATP7B* gene (encodes ATPase copper transporting-β), display increased sensitivity to copper treatment, owing to defective copper transport and toxic intracellular accumulation[58]. Although the *Atp7b*-knockout mouse model captures certain hepatic features of the disease, such as copper overload, inflammation and fibrosis, it fails to reproduce the neurological and psychiatric manifestations seen in patients[59]. Liver organoids derived from patients with biliary atresia show aberrant morphology, characterized by multiple vacuoles, reduced proliferation ability, disrupted apical–basal organization and low levels of CFTR protein[60,61]. These phenotypes are consistent with obstructive bile duct disease in patients, and transcriptome analysis has further revealed amyloid-β (Aβ) deposition around bile ducts as a novel pathological and diagnostic feature[61].

Genetic disorders of the CNS. Parkinson disease (PD) is characterized by the loss of midbrain dopamine neurons, leading to both motor and non-motor symptoms. PD symptomatic neurodegenerative phenotypes, as well as other CNS disorders with generally later onset, remains challenging to model using organoids given the fetal maturation state of iPS cell-derived dopaminergic neurons. Therefore, most efforts have focused on recapitulating early pathological processes with organoids carrying disease-associated mutations, which can reveal mutation-specific disease mechanisms. For example, human ES cell-derived midbrain organoids with a loss-of-function mutation in *DNAJC6*, a gene linked to early onset of PD, display impaired dopamine neuron development, pathological α-synuclein aggregation, increase in intrinsic neuronal firing frequency, and mitochondrial and lysosomal dysfunctions, due to impaired WNT–LMX1A signalling[62]. Notably, *Dnajc6*-knockout mice do not recapitulate PD-associated phenotypes[62]. Similarly, iPS cell-derived midbrain organoids carrying the PD-associated leucine-rich repeat serine/threonine protein kinase 2 (*LRRK2*) G2019S mutation recapitulate hallmarks observed in patients, including increased aggregation of α-synuclein and impaired clearance. Further analysis has identified TXNIP as a key factor in *LRRK2*-associated PD[63].

In another example involving a common neurodegenerative disease, the apolipoprotein E variant *APOE4* is the strongest genetic risk factor for Alzheimer disease (AD). iPS cell-derived cerebral organoids carrying an *APOE4* variant show enhanced AD pathology, with increased levels of Aβ and phosphorylated tau, cell apoptosis and reduced synaptic integrity[64]. Notably, besides the direct effects of the *APOE4* variant on neurons, a co-culture organoid model involving neurons, astrocytes and microglia has revealed the role of microglia *APOE4* in AD pathology by inducing neuronal inflammation[65]. Further transcriptional profiling has identified cell-type-specific changes associated with the *APOE4* variant, including impaired synaptic function in neurons, decreased Aβ uptake and cholesterol accumulation in astrocytes, and altered morphology and reduced Aβ phagocytosis in microglia-like cells, respectively[66]. Organoids derived from iPS cells of patients with familial AD (harbouring duplication of *APP*, which encodes amyloid-β precursor protein, or mutations in *PSEN1* or *PSEN2*, which encode presenilin 1 or 2, respectively) develop pathologies spontaneously – such as amyloid aggregation, hyperphosphorylated tau and endosome abnormalities – which can be rescued by β- and γ-secretase inhibitors[67].

Although hyperphosphorylated tau is a hallmark of AD, mutations in *MAPT*, which encodes tau, can cause frontotemporal dementia (FTD), a heterogeneous disease characterized by tau accumulation and loss of glutamatergic cortical neurons. iPS cell-derived cerebral organoids carrying a *MAPT* mutation (R406W) have been used to study the specific mechanisms of tau pathology. The R406W mutation induces tau mislocalization and axonal dystrophy via microtubule destabilization, which also disrupts mitochondrial transport in mutant neurons[68]. The mutant tau protein has reduced phosphorylation and is more susceptible to calpain-mediated cleavage and fragmentation. Similarly, organoids harbouring another *MAPT* mutation (V337M) have been used to investigate early FTD pathogenesis[69]. These organoids are phenotypically assessed at multiple time points to identify various steps in the development of FTD-tau pathology. FTD hallmarks, including neuronal loss, progressive tau accumulation and susceptibility to glutamate toxicity, are observed at later stages (after 4 months). The accumulation of tau is primarily due to defects in proteolytic activity, which is caused by disruption of the autophagy–lysosomal pathway, rather than to increased tau expression. Further gene enrichment analysis revealed increased expression of *ELAVL4*, a splicing regulator gene that encodes a neuronal RNA-binding protein, leading to splicing dysregulation and impaired excitatory neuron function. This dysregulation of excitatory neuron development results in excitotoxicity – a process whereby overstimulation by excitatory neurotransmitters, primarily glutamate, leads to excessive influx of calcium ions into neurons, causing cellular dysfunction and eventual death – and apoptosis in the late stages of development. Additionally, the loss of glutamatergic neurons in mutant-tau organoids can be rescued by a lipid kinase inhibitor, apilimod, which prevents recycling of glutamate receptor to the membrane, thereby reducing the glutamate-induced excitotoxicity[69]. An alternative approach using adeno-associated virus (AAV) injection of mutant tau (P301L) into the iPS cell-derived forebrain organoids also induces pronounced tau aggregates containing tau fibrils[70]. Besides *MAPT* mutations, FTD has various other genetic causes, including *GRN* mutations that lead to haploinsufficiency of progranulin protein and GGGGCC repeat expansion in *C9ORF72*. A cerebral organoid slice model derived from iPS cells recapitulates cortical architecture and displays early molecular pathology of *C9ORF72*-associated FTD. In these organoids, astroglia show increased levels of the autophagy-related protein P62, while deep layer neurons accumulate dipeptide repeat protein poly(GA), DNA damage and undergo nuclear pyknosis that can be pharmacologically rescued by GSK2606414 (ref. 71). iPS cell-derived FTD organoids with *C9ORF72* expansion also show exacerbated TDP43 dysfunction, a key driver of acute injury. A CRISPR interference (CRISPRi) screen based on FTD organoids has identified KCNJ2 inhibition as a promising therapeutic target to mitigate neurodegenerative processes[72].

Neurodevelopmental disorders can also be modelled in organoids. Fragile X syndrome (FXS) is caused by the loss of fragile X mental retardation protein (FMRP), an RNA-binding protein that regulates the translation of specific mRNAs that are involved in synaptic development and plasticity, including brain-derived

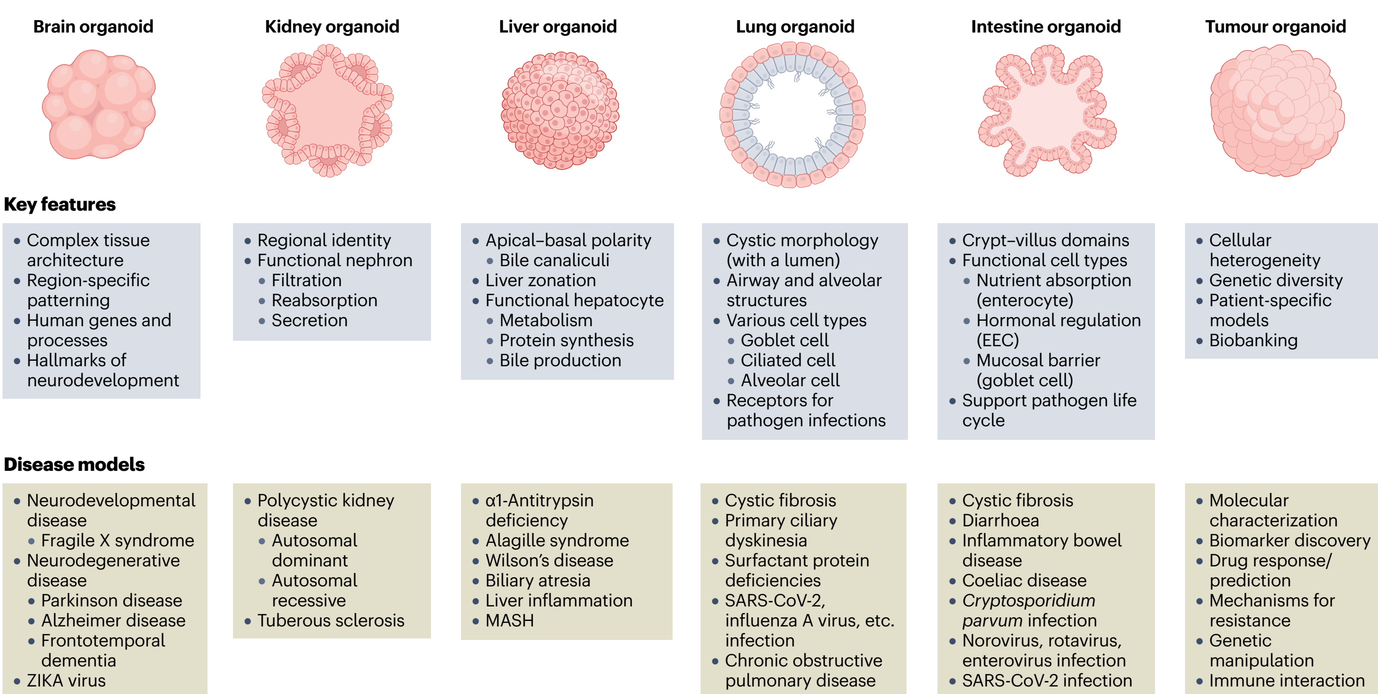

Fig. 2 | Different types of organoid in disease modelling. Organoids replicate the (patho-)physiological characteristics of real organs, such as the brain, kidney, liver, lung and intestine. Compared with immortalized cell lines, organoids exhibit complex tissue architecture with regional identity and correct cellular polarity. They contain mature cell lineages that facilitate functional assays. Notably, organoids express essential receptors and intracellular elements that enable pathogen infections and support the complete life cycles of these pathogens. These unique features, summarized in the upper panels, make organoids valuable for modelling diseases, including genetic disorders, infections and various forms of cancer. Furthermore, by incorporating niche components such as immune cells, organoids can model and investigate inflammatory diseases, and provide insights into disease mechanisms and potential treatments. Examples of the modelled diseases are displayed in the bottom panels. MASH, metabolic dysfunction-associated steatohepatitis; EEC, enteroendocrine cell; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

Box 1 | Organoid evolution

Most current organoid models lack crucial tissue components such as vasculature, immune cells and innervation, which are essential to recapitulate the full physiological environment of human tissues. Researchers have begun to incorporate vascular networks into organoid models by co-culturing organoids with endothelial cells ('3D bioprinting' or 'micro-engineered assembloids') or using microfluidic systems ('organoid-on-a-chip') to create channels for blood flow (Fig. 3). For example, the addition of human endothelial cells to liver organoids has resulted in the formation of functional blood vessel-like structures that improve tissue function and mimic the liver's natural perfusion[282–285]. Similarly, the absence of immune cells in organoids has been addressed by incorporating immune cell populations, such as macrophages[286], dendritic cells[287] and T cells[183], into organoid cultures. One approach used in cancer research is the integration of patient-derived immune cells into tumour organoids, which enables the study of immune responses in the tumour microenvironment and testing of immune-modulating therapies[151]. Lastly, the lack of innervation in organoids has been partially overcome by co-culturing organoids with neurons or incorporating neural progenitor cells to create neuro-organoids[288,289].

For relatively homogeneous organoids or assays that do not depend on 3D architecture, culturing organoid cells as a 2D monolayer is a viable alternative. The starting cell mass is typically generated in 3D culture. The 2D approach reduces inter-well variability, improving assay consistency and reproducibility. The flat monolayer also supports fast and high-content imaging and facilitates integration into complex systems such as 'organoid-on-a-chip', which investigate drug effects in the presence of stromal cell types, including endothelium and immune cells. We have recently described a bacterial protein, *Yersinia* invasin, which can be coated on 2D transwells and fully replaces Matrigel[290]. Notably, the 2D transwell system can also improve drug safety and metabolic testing, which may require accessibility to both apical and basal surfaces of the organoids.

neurotrophic factor (BDNF), metabotropic glutamate receptors (mGluRs), calcium/calmodulin-dependent protein kinase II (CaMKII) and postsynaptic density protein 95 (PSD95). In organoids derived from forebrains from patients with FXS, loss of FMRP leads to dysregulated neurogenesis, neuronal maturation and excitability[73]. These deficits can be rescued by inhibiting the phosphoinositide 3-kinase (PI3K) pathway, but not by the mGluR5 antagonist[73]. Of note, mGluR5-targeted therapy has shown limited success in clinical trial, but promising results in an animal model[74]. Transcriptome analysis of these FXS organoids further revealed human FMRP mRNA targets, such as *CDH2*, providing mechanistic insights and potential druggable targets for the therapeutic development of FXS. A notable limitation of this model is the considerable reduction of GABAergic inhibitory neurons in FXS organoids. Therefore, further investigation into FMRP loss in the development of GABAergic neurons would benefit from another model system enriched in GABAergic neurons, such as ventral forebrain organoids. Additionally, this FXS organoid model cannot fully capture the diversity of cell types and their complex interactions, especially those that involve astrocytes, which are reported to mediate cell non-autonomous correction of aberrant firing in human FXS neurons[75]. A recent study suggests that FMRP can inhibit neuronal translation mediated through YTH domain-containing family protein 1 (YTHDF1). In an iPS cell-derived forebrain organoid model, a small-molecule inhibitor targeting YTHDF1 reversed developmental defects associated with FMRP deficiency[76].

Metabolic disorders

Metabolic dysfunction-associated steatohepatitis (MASH), in which lipid accumulation in hepatocytes drives chronic inflammation and fibrosis, affects approximately 5% of the global adult population[77]. Risk factors for MASH include nutritional overload or, much more rarely, genetic lipid disorders[78]. MASH animal models driven by either genetic changes (mutations in leptin or leptin receptor or knockout of genes associated with lipid metabolism such as *Apoa5*, *Nr1h4*, *Pparg*, *Cebpa*) or by dietary factors (high-fat diet, Western diet or excessive sugar intake) are widely used. However, these animal models are not suitable for large-scale drug screening and, in addition, interspecies differences complicate the translation of findings to human physiology.

Hepatocyte organoids represent the 3D architecture of liver tissues and recapitulate the gene expression profiles and cellular functions of primary hepatocytes[79]. MASH organoids derived from patients[80] or induced by metabolite challenge[81] capture essential disease features, characterized by upregulation of pro-inflammation pathways and tumour-associated markers, increased lipid accumulation and sensitivity to apoptosis. For example, a hepatic organoid system that shows lipid accumulation upon free fatty acid (FFA) treatment has been used to study the effects of metformin and L-carnitine, both of which are used in the treatment of type 2 diabetes (T2D)[82]. Liver organoids derived from patients with Wolman disease – caused by a genetic dysfunction of lysosomal acid lipase (LAL) that leads to impaired triglyceride hydrolysis – show massive lipid accumulation and severe steatohepatitis. These phenotypes can be rescued by treatment with fibroblast growth factor 19 (FGF19), which activates farnesoid X receptor (FXR) to reduce lipid storage-induced reactive oxygen species (ROS) activity[83]. Other compounds have also been evaluated in liver-related organoid models, with mixed results. Magnesium – although reported to improve T2D and liver metabolic disorders[84] – is ineffective in reversing the disease phenotype in 3D organoid models[83]. By contrast, sorafenib, a broad-spectrum tyrosine kinase inhibitor used clinically in the treatment of liver and kidney cancers, has shown potential in reducing steatosis-induced fibrogenesis in 3D co-culture models of MASH[85].

Notably, the interplay between inflammatory changes and fibrotic progression in MASH involves interactions with Kupffer cells (KCs) and hepatic stellate cells (HSCs). Multi-cell-type liver organoids derived from iPS cells have been used to investigate MASH pathogenesis and facilitate drug screening[83,86]. In this system, FFAs stimulate pro-collagen production[86]. Prolonged fatty acid exposure results in increased expression of inflammatory cytokines in KCs, activation of HSCs and elevated organoid stiffness[83]. These findings underscore the importance of generating multilineage interactions for improved MASH modelling.

Organoids have not only modelled metabolic disease but have also facilitated new therapeutic discoveries. iPS cell-derived hepatocytes from patients with familial hypercholesterolaemia have revealed the effect of cardiac glycosides in reducing apolipoprotein B production[87]. Hepatocytes derived from patients with mitochondrial DNA depletion syndrome have demonstrated the improvement of mitochondrial function by NAD treatment, which restores ATP levels through activation of peroxisome proliferator-activated receptor-γ coactivator 1α (PGC1α)[88]. Hepatocyte organoids are valuable for screening small-molecule inhibitors or conducting CRISPR-based functional studies in the context of preclinical metabolic dysfunction-associated steatotic liver disease (MASLD) and MASH[89]. We have used fetal hepatocyte organoids to model the early onset of MASLD, in the context of genetic steatosis (*APOB*$^{-/-}$ or *MTTP*$^{-/-}$, the genes encoding apolipoprotein B and microsomal triglyceride transfer protein, respectively) or an overload of FFA[90]. As a proof of concept, both conventional drug screening and CRISPR-mediated loss-of-function screening can be carried out in these MASLD organoids, highlighting *FADS2* (encodes fatty acid desaturase 2) as one determinant of hepatic steatosis[90].

Chronic inflammatory disease

Modelling of immune-mediated diseases, such as inflammatory bowel disease (IBD) or coeliac disease (CeD), requires the combination of organoids with the appropriate immune cells (Box 1) and/or inflammatory mediators.

Numerous risk factors have been identified through genome and transcriptome analysis of patients with IBD[91,92]. However, elucidation of their major cellular targets and underlying mechanisms has proved challenging in animal models. Organoids derived from patients with IBD[93] or exposed to pro-inflammatory cytokines[94] allow for straightforward investigations. For instance, analysis of IL-22-stimulated colon organoids has revealed upregulation of genes involved in recruiting C-X-C chemokine receptor type 2 (CXCR2)$^{+}$ neutrophils[95]. Of note, organoids derived from IBD-inflamed mucosa retain transcriptional signatures of inflamed tissues[96], and disease-specific epigenetic alterations in paediatric patients with IBD can also persist in organoids even after removal of the inflammatory environment[97]. A physiologically relevant

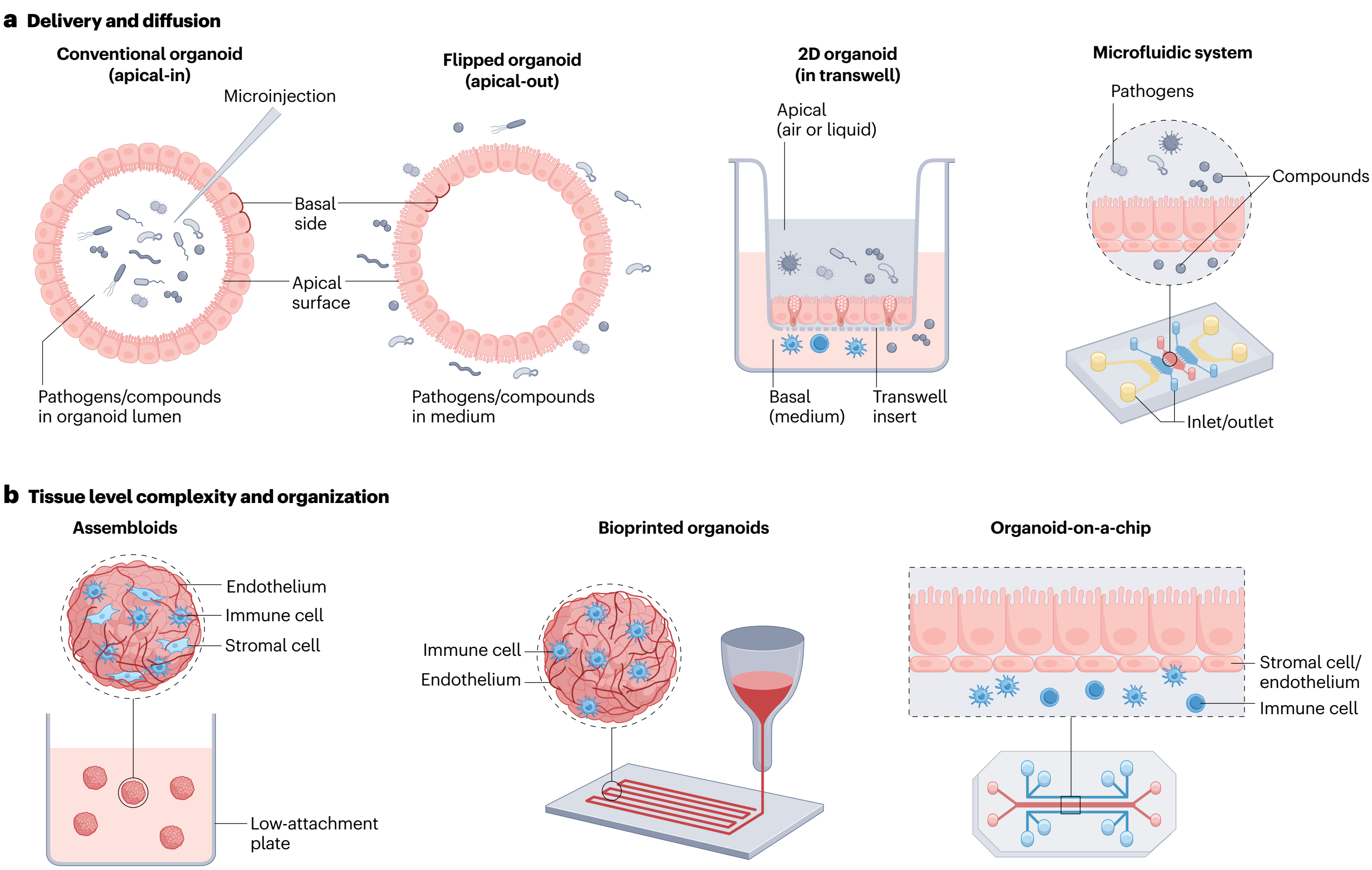

Fig. 3 | Advanced organoid systems in disease modelling and drug safety evaluation. a, Strategies for substance delivery and diffusion. Pathogen receptors and compound transporters or channels can be asymmetrically distributed on the apical surface of the cells. Conventional organoids, typically embedded in Matrigel, expose their basolateral surface (apical-in) and require microinjection of pathogens or compounds into the organoid lumen to improve delivery efficacy. Flipped (apical-out) organoids, generated through the removal of Matrigel, mitigate challenges related to delivery and diffusion while preserving the 3D architecture of the organoids. In 2D transwell cultures, both the apical and basolateral surfaces of organoid cells are readily accessible, allowing for comprehensive interaction with external environments. Similar to the transwell systems, bioengineered microfluidic platforms offer precise control over substance delivery from both the apical and basolateral surfaces of organoid cells via multiple chambers equipped with inlets and outlets. **b**, Advanced organoid models with improved tissue-level complexity and organization. In a Matrigel-free suspension culture system, a mixture of various cell types can spontaneously aggregate to form assembloids, with the extracellular matrix provided by different cell types such as endothelial and stromal cells. Although assembloids enhance cellular complexity substantially, they fall short of accurately mirroring the tissue organization. Bioprinted organoids exhibit superior arrangement of diverse cell lineages, closely resembling the architecture of natural tissues, that is, with improved vascular networks, innervation patterns and zonation. When cultured on a bioengineered chip, different cell types (such as endothelial, stromal or immune cells) can be systemically incorporated and organized, using tailored designs to achieve a sophisticated tissue-level architecture.

method to study IBD pathogenesis involves organoid–immune cell co-culture systems. For example, autologous mucosal T cells from patients with Crohn's disease can infiltrate organoids and directly induce epithelial cell death. This infiltration and cytotoxicity depend on T cell recognition through CD103 (an integrin that mediates T cell adhesion to epithelial cells) and NKG2D (an activating receptor that binds to stress-induced ligands on epithelial cells). Blocking antibodies to CD103 and NKG2D target immune cell–epithelium interactions to reduce T cell infiltration and epithelial damage[98].

CeD is an autoimmune disorder triggered by gluten and associated with specific HLA-DQ haplotypes[99]. Mice do not naturally possess these human HLA haplotypes and, although transgenic mice expressing human HLA molecules have been developed, they do not fully replicate the CeD immune response or disease pathology[100] or they require artificial induction or genetic modifications that do not fully replicate the natural disease process[101]. Duodenal biopsy samples from patients with CeD can be directly cultured in an air–liquid interface system (Fig. 3a), which preserves the gut epithelium along with diverse tissue-resident immune populations, thereby recapitulating gluten-dependent pathology[102]. Functional studies with this co-culture system have identified IL-7 as a gluten-inducible pathogenic modulator that is necessary and sufficient to induce epithelial destruction in CeD pathogenesis[102].

The thyroid gland is a well-known target of organ-specific autoimmunity: hyperthyroidism Graves' disease and hypothyroidism Hashimoto thyroiditis represent autoimmune conditions characterized by aberrant hormone production[103]. Tissue-derived thyroid organoids containing hormone-producing follicular cells can readily be generated[104]. Thyroid organoids exposed to Graves' disease auto-antibodies (known to activate the thyroid-stimulating hormone receptor, TSHR) show increased cell proliferation and hormone secretion, reproducing key disease features[104]. Organoids derived from patients with Hashimoto thyroiditis showed elevated levels of chemokines CCL2 and CCL3, which may have crucial roles in recruiting immune cells during Hashimoto thyroiditis pathogenesis[105].

Chronic obstructive pulmonary disease (COPD) primarily results from damage to alveoli and airways, leading to reduced airflow, increased mucus production and inflammation, which collectively contribute to breathing difficulties. Bronchial organoids from patients with COPD confirm the presence of goblet cell hyperplasia and reduced ciliary beat frequency[106]. Smoking-related COPD has been investigated by subjecting human lung organoids to cigarette smoke extract, which results in suppressed organoid formation and increased susceptibility of alveolar organoids to cigarette smoke compared with healthy airway organoids[107].

Infectious diseases

Cryptosporidium parvum, a diarrhoea-causing protozoan parasite, has a complex asexual–sexual life cycle and has resisted efforts to be propagated in the lab. Human intestinal organoids, particularly differentiated enterocytes, support both the asexual and sexual stages of the *Cryptosporidium* life cycle and produce oocysts with secondary infection capabilities comparable to those derived from infected host animals[108].

Similarly, human intestinal organoids have provided the first in vitro human norovirus (HuNoV) platform, identifying enterocytes as the target for replication[109]. HuNoV infections are the leading cause of acute gastroenteritis worldwide, yet traditional cell lines fail to support HuNoV infection and replication, and mice are naturally resistant to infection owing to species-specific differences in viral receptors and immune responses[110]. Furthermore, murine norovirus models differ from HuNoV in terms of viral structure, receptor usage and pathogenicity, limiting the direct translation of the findings to human disease[110]. Intestinal organoids overcome these barriers, displaying strain-dependent HuNoV binding specificity for histo-blood group antigens[111], consistent with findings from animal and human studies[112,113]. Beyond HuNoV, intestinal organoids have been applied to model other enteric virus infections, including rotavirus[114,115], coxsackie B virus[116] and enterovirus[117,118].

In 2016, the WHO declared a public health emergency of international concern about the association of ZIKA virus (ZIKV) infection with clusters of neonatal microcephaly and other neurological disorders in Brazil[119]. Lack of access to live infected human fetal tissues limited the understanding of ZIKV pathogenesis in the developing CNS. The microcephaly phenotypes require a 3D architectural model that can only be recapitulated through infection of brain organoids[120–122]. Experiments using human iPS cell-derived cerebral organoids have shown that ZIKV infection disrupts cerebral cortical layers, decreases cell proliferation and reduces the number of functional neurons[120]. Region-specific brain organoids grown in spinning bioreactors have demonstrated that both Asian and African ZIKV strains preferentially and productively infect neural progenitors, triggering premature differentiation and increased cell death, mirroring infection patterns observed in human fetuses[121,123,124]. Progenitors in the ventricular zone and subventricular zone can be affected, supporting the hypothesis that proliferating progenitor cells during human neurodevelopment are the key target in the disease[124]. Mechanistic studies based on these organoid models revealed that ZIKV protein NS2A impairs proliferation of radial glial cells in human forebrain organoids and disrupts apical junction formation in neural stem cells[125]. The expression of ZIKV proteins NS4A and NS4B blocks neurogenesis and promotes autophagy by inhibiting AKT–mTOR signalling[126]. ZIKV protein NS5 specifically targets human, but not mouse, STAT2 to inhibit type I interferon signalling[127], and ZIKV can activate Toll-like receptor 3 (TLR3)-mediated innate immune response, leading to dysregulation of genes involved in neurogenesis, axon guidance and apoptosis[128]. Human forebrain organoids have also been adopted in drug screening, identifying hippeastrine hydrobromide as an agent that eliminates ZIKV in infected human neural progenitors and rescues ZIKV-induced microcephaly phenotype[129]. Another drug-repurposing screen has found that emricasan, a pan-caspase inhibitor that suppresses ZIKV-induced caspase 3 activity, can protect human cortical neural progenitors in organoid cultures, and an FDA-approved category B anthelmintic drug niclosamide can block ZIKV replication[130]. Furthermore, ZIKV infection increases transient receptor potential channel 4 (TRPC4) expression in host cells via interaction between the viral NS3 protein and CaMKII, which enhances TRPC4-mediated calcium influx. Pharmacological inhibition of CaMKII or TRPC4 can potentially reduce seizures and block ZIKV spread in iPS cell-derived brain organoids[131].

Respiratory infections. The respiratory tract represents a major target for pathogens, and the coronavirus disease 2019 (COVID-19) pandemic has prompted an unprecedented need for in vitro models. Airway ciliated cells expressing angiotensin-converting enzyme 2 (ACE2) and transmembrane protease serine 2 (TMPRSS2) represent the primary cellular targets during initial infection in differentiated airway organoids[132]. Given that ACE2 and TMPRSS2 are expressed across various epithelial tissues, severe acute respiratory syndrome coronavirus 2

(SARS-CoV-2) can disseminate to other organs following initial infection in the respiratory tract. Human intestinal organoids containing ACE2-expressing enterocytes enable robust SARS-CoV-2 infection and replication[133]. Follow-up studies using CRISPR-mediated genetically engineered organoid biobank demonstrated that SARS-CoV-2, as well as Middle East respiratory syndrome coronavirus (MERS-CoV) and SARS-CoV, specifically use TMPRSS2, which proteolytically activates the spike protein, allowing direct fusion of the viral envelope with the host cell membrane. Thus, these viruses infect host cells without requiring endocytosis, making TMPRSS2 an attractive target for pan-coronavirus therapeutics[134]. Of note, (hydroxy)chloroquine, a blocker of endocytosis, was originally proposed based on cell line studies and was widely prescribed for patients with COVID-19. Yet, this drug does not inhibit viral reproduction in organoid models[134]. This finding was consistent with clinical observation that (hydroxy) chloroquine is ineffective against SARS-CoV-2 (ref. 135), highlighting the potential of organoids as a more accurate translational platform for screening for novel drug treatments.

More broadly, airway organoids can be used to study the infection dynamics and zoonotic spillover potential of respiratory viruses such as influenza A virus (IAV), respiratory syncytial virus (RSV) or human rhinovirus C (HRV-C). They express TMPRSS2 and TMPRSS4, which mediate cleavage of the IAV surface haemagglutinin, thus enabling virus–host membrane fusion[136]. Differentiated 2D organoids (Box 1, see also Fig. 3a) can distinguish the replication capacities of human-infective IAVs from poorly human-infective avian and swine subtypes[136] and have been used to assess the spillover risk of recent avian H5N6 and H5N8 strains[137]. Long-term expandable human airway organoids also recapitulate key features of RSV infection, including increased cell motility driven by the non-structural viral NS2 protein and the preferential recruitment of neutrophils during co-culture[43]. Similarly, they can sustain reproducible HRV-C propagation, providing insights into virus–host interactions[138]. Following HRV-C infection, airway organoids mount a stronger innate immune response than nasal organoids, and treatment with α-CDHR3 and antivirals considerably reduces viral growth[138], indicating their potential as a platform for antiviral drug development.

Cancer

Many forms of cancer – such as pancreatic, liver and breast cancer – do not develop spontaneously in animal models. When tumours are experimentally induced by introducing specific cancer mutations (which often requires considerable time), they still fail to capture the complexity and heterogeneity of human cancers, including their genetic diversity and immune environment. Over the past decade, the emergence of patient-derived tumour organoids (PDTOs) has greatly expanded the repertoire of preclinical cancer models[139]. These cultures can be readily established from virtually all carcinomas, using biopsy or resection material as starting tissue. They are scalable and closely recapitulate the histological, genetic and transcriptomic features of original tumours[140], and they can consistently be xenotransplanted into mice for in vivo characterization[141]. PDTOs carrying specific genetic mutations can grow independently of the growth factors that signal through the mutationally activated oncogenic pathways. For example, organoids with the *KRAS*^{G12V} mutation can grow in the absence of epidermal growth factor (EGF), which is essential for the growth of healthy organoids[142], an approach commonly used to distinguish tumour-derived outgrowth from normal tissue growth[143]. The utility of tumour organoids has been further demonstrated in drug testing studies that showed mutation-specific cytotoxic effects. For example, sensitivity to afatinib, an irreversible inhibitor of the ERBB family of receptor tyrosine kinases, correlates with HER2 expression levels in breast cancer organoids. Similarly, organoids with mutant *BRCA1/2* signatures are more sensitive to poly(ADP-ribose) polymerase (PARP) inhibitors, leading to tumour cell death[144]. In addition, defined cancer models can be created by sequential CRISPR engineering of oncogenic mutations into wild-type organoids[142]. Such artificially created cancer organoids yield the expected histology upon xenotransplantation[142].

PDTOs, along with their matching healthy organoids, have been extensively generated and stored in biobanks across various tumour types (summarized in Table 2 and Supplementary Table 1). These PDTOs reflect both inter- and intra-tumour heterogeneity, allowing for the study of tumour evolution and progression, testing the variability of treatment response, understanding resistance mechanisms and identifying novel therapeutic targets[145,146]. As an example, MCLA-158 (petosemtamab) is a bispecific antibody directed towards EGFR and LGR5 that was developed entirely using colon cancer PDO lines (Box 2). Importantly, PDTOs display sensitivity to antitumour therapies comparable to that of their parental tumour tissues, underscoring the reliability of PDTOs in drug screening[147–150]. Of note, in their 'ground state', PDTO lines do not contain immune cells, tumour stroma or blood vessels, yet these elements can be readily included by co-culturing[151–153], supported by microfluidic systems and bioengineering technologies such as organoid-on-a-chip and 3D bioprinting (Box 1, see also Fig. 3b).

Despite these advantages, organoids represent a highly reductionist experimental platform, and although readouts tend to be robust and reproducible, outcomes must be validated extensively against existing in vitro platforms, animal studies and clinical trial data before organoids can be applied routinely in drug development. PDTOs show promise for advancing personalized medicine across multiple carcinoma types, with correlations between organoid drug responses and clinical outcomes in pancreatic[154], gastrointestinal[145,155] and breast[144] cancers. However, their broader translatability remains to be conclusively demonstrated, and multiple randomized clinical trials are currently ongoing[156–160] to assess their predictive value, particularly in radiotherapy and chemotherapy settings. Importantly, outside of oncology, translational evidence remains scarce, and controlled clinical validation is largely lacking. In addition, negative predictive data are seldom reported, limiting our understanding of the boundaries of organoid-based prediction. In neurological and other complex systemic diseases, organoids are further challenged by the difficulty of modelling multicellular interactions and systemic pharmacokinetics, which constrains their predictive utility.

Translational safety applications

Although most studies have focused on organoids for drug efficacy and disease modelling, their use is now expanding within the pharmaceutical industry to assess the safety of pharmacological compounds across multiple organs (Table 3).

Liver toxicity

The liver has a central role in drug metabolism, and hepatotoxicity is a leading cause of drug attrition, market withdrawal and restricted use after approval[161]. Drug-induced liver injury (DILI) represents an important treatment-associated cause of morbidity and mortality and remains challenging to predict and mitigate[162]. Liver organoids, derived from either TSCs or PS cells, represent a promising opportunity for prediction of drug-induced hepatotoxicity.

Table 2 | Representative PDTO biobanks for drug screening and assessment

Tissue	No. of PDTO lines	Panel of drugs or treatments	Key findings	Clinical comparison	Ref.
Gastric	46	37 anticancer drugs that are FDA approved or in clinical trials	Effective drugs including napabucasin, abemaciclib and ATR inhibitor VE-822	Similar response to 5-FU and cisplatin between organoids and patients	145
Colorectal	22	83 drugs in clinical use, clinical trials, chemotherapies and experimental stages	Genetic correlates between individual oncogenic mutations and drug responses	NC	140
	65	5-FU, FOLFOX, radiation	PDTOs display heterogeneous sensitivity to chemotherapies observed clinically	PDTOs reveal a diversity of responses to chemotherapies and radiation that correlate with clinical response	155
	19	87 chemotherapeutic drugs and targeted agents	Functional drug sensitivities are consistent with genomically predicted targets	Response in organoid-guided therapy choices for this poor-prognosis cohort	275
Pancreas	103	5 chemotherapeutic drugs and 21 targeted agents	PDTO-derived gemcitabine-sensitivity signature stratifies patients with pancreatic cancer with improved response to adjuvant gemcitabine	PDTO pharmacotyping corresponds to individual patient treatment responses	154
Liver	8	29 anticancer compounds, including drugs in clinical use or development	ERK inhibitor SCH772984 as a potential therapeutic agent for primary liver cancer	NC	276
Breast	95	Drugs targeting the HER signalling pathway	HER2-overexpressing organoids are sensitive to drugs targeting the HER pathway	PDTOs recapitulate the in vivo drug responses of their original tumours	144
Ovary	36	17 chemotherapeutic drugs and targeted agents	Interpatient drug response heterogeneity correlates partially with their genetic make-up	Carboplatin and paclitaxel treatment in PDTOs exhibit significant correlation with the clinical response	150
Head and neck	26	Cisplatin, carboplatin, cetuximab, inhibitors targeting BRAF, PARP, mTOR, FGFR, and radiotherapy	PDTOs exhibit differential responses to a panel of clinical drugs and show selective sensitivity to the targeted agents	Correlation of in vitro PDTO responses with clinical responses in 7 patients who received radiotherapy	277
Oesophagus	16	Chemotherapeutic drugs and targeted agents	PDTOs derived from endoscopic biopsy recapitulate the pre-treatment tumour	Drug treatment response of PDTOs reveals similarity to tumour response	278
	10	24 FDA-approved drugs and preclinical agents	Notable sensitivities to MEK1/2 inhibitor trametinib and EGFR/HER2 inhibitor afatinib	The lack of chemotherapy sensitivity of most organoid cultures is consistent with the poor in vivo response	279
Kidney	33	24 chemotherapeutic drugs and targeted agents, CD19 and CD79 CAR-T cells	Effective drugs are AKT inhibitor MK-2206, MEK1/2 inhibitor trametinib and ERK1/2 inhibitor SCH772984, but not RTK inhibitors; CD70 CAR-T cells exhibit tumour-specific response	Consistent with their in vivo response, PDTOs were resistant to conventional chemotherapeutic drugs	280
Brain	53	Drugs targeting EGFR–mTOR–MEK pathway, EGFRvIII-specific CAR-T cells	Mutational profiles correct to specific drug responses; specific CAR-T cell response to EGFRvIII-expressing organoids	NC	146
	1	Genetic mutation-guided drug candidates	PDTOs predict drug responses in glioblastoma	Prospective identification of everolimus as a treatment option using PDTOs showing subsequent partial clinical response	281

A full version of this table is provided in Supplementary Table S1. CAR, chimeric antigen receptor; 5-FU, 5-fluorouracil; EGFR, epidermal growth factor receptor; FGFR, fibroblast growth factor receptor; FOLFOX, folinic acid, fluorouracil and oxaliplatin; NC, no comparison; PARP, poly(ADP-ribose) polymerase; PDTO, patient-derived tumour organoid.

Functionally mature human liver organoids (HLOs) from PS cells can recapitulate the known hepatotoxicity of troglitazone (an anti-diabetes drug withdrawn for liver toxicity), paracetamol (also known as acetaminophen; a widely used analgesic) and the antibiotics trevafloxacin and levofloxacin, with a higher sensitivity than 2D hepatocytes at clinically relevant concentrations[82]. Similarly, TSC-derived HLOs outperform conventional hepatocellular carcinoma-derived HepG2 cells in modelling drug-induced phospholipidosis, a severe adverse effect associated with a wide range of drugs, highlighting their capacity to evaluate metabolite-mediated hepatotoxicity[163]. Moreover, PS cell-derived HLOs have been used to establish a high-throughput assay for large-scale compound screening, accurately annotating the DILI potential of 238 drugs with multiplexed readouts and demonstrating their ability to model genotype-specific susceptibility to bosentan — an endothelin receptor antagonist that can cause cholestatic liver injury — in individual donors[164]. Building on these advances, HLOs can model a range of DILI-associated phenotypes, including steatosis, fibrosis and immune responses, with strong concordance to clinical data using compounds such as paracetamol and TAK-875, a G protein-coupled receptor 40 (GPR40) agonist formerly developed for T2D but discontinued owing to hepatotoxicity[165]. Expanding HLO applications, intrahepatic cholangiocyte organoids have been used to study chlorpromazine-induced bile duct injury, and have demonstrated that this model effectively replicated drug-induced biliary toxicity[166]. More recently, HLOs have been adapted for the evaluation of AAV vectors, which are widely used in gene therapies, particularly for liver-targeted treatments, providing a novel platform to assess vector performance and on-target hepatoxicity in gene therapy development[167].

Kidney toxicity

Human PS cell-derived kidney organoids, although still lacking several key transporters and tubular filtration, can model drug-induced acute kidney injury, a major safety concern in drug development[168]. These organoids contain nephrons, collecting ducts, renal interstitium and endothelial cells, and can self-organize into nephron-like structures, including podocytes, proximal tubules, loops of Henle and distal tubules, further recapitulating in vivo nephron organization[18]. They have demonstrated differential apoptosis in response to nephrotoxicants such as cisplatin and gentamicin, with dose-dependent upregulation of the kidney injury marker kidney injury molecule 1 (KIM1), establishing their utility for nephrotoxicity testing[169,170]. Subsequent studies have refined kidney organoid models to better replicate nephrotoxic responses. Cisplatin and doxorubicin-induced injury[171,172], including dose-dependent apoptosis[173], has been reproduced in conventional organoids, and micro-organoids in suspension culture, and bioprinted kidney organoids have demonstrated toxicity from doxorubicin and aminoglycoside antibiotics[174,175]. An immune-competent, vascularized kidney organoid-on-a-chip system incorporating perfused peripheral blood mononuclear cells (PBMCs) has further enabled the assessment of immune-mediated toxicity, exemplified by the selective killing of WT1-expressing cells in response to a T cell bispecific antibody (TCB) targeting an HLA-A2-presented WT1 peptide[176]. This highlights the capacity of this system to evaluate both chemical and immune-mediated toxicities in a dynamic microenvironment.

Gastrointestinal toxicity

Better prediction of gastrointestinal toxicity with organoid models for both small and large molecules could improve preclinical risk–benefit assessment considerably and reduce attrition during drug development.

A pioneering study has used human ileal TSC-derived organoids to model drug-induced diarrhoea, establishing an organoid-based assay to validate against 31 reference drugs. This assay demonstrates 90% accuracy in predicting the clinical incidence of drug-induced diarrhoea[177]. Building on this, the molecular determinants of gastrointestinal toxicity have been explored using TSC-derived small intestine and colon organoids, investigating mechanisms of doxorubicin- and gefitinib-induced side effects[178,179]. A cost-effective method of intestinal organoid culture has been developed, enabling phenotypic HTS of small molecules, helping to streamline preclinical work-flows[180]. Using this platform, undifferentiated human organoids can accurately distinguish between compounds with cytotoxic or non-cytotoxic potential. Moreover, comparing compound responses in human, rat and dog organoids revealed species-specific sensitivity differences that align with preclinical findings and demonstrated the utility of organoids for early safety assessment[181]. The predictive capabilities of intestinal organoid platforms extend to the assessment of 'on-target, off-tumour' effects of protein-based biologics such as TCBs, including the lack of cross-reactive preclinical species. Intestinal PDOs and tumouroids supplemented with immune cells have shown tissue-specific off-tumour toxicities of TCBs, successfully capturing clinical adverse effects that are not predicted by conventional tissue-based models that lack these immune components[182]. Expanding this approach, human intestinal immuno-organoids are generated by integrating epithelial organoids with autologous tissue-resident memory (TRM) T cells, and they have been used to investigate TCB-induced intestinal inflammation observed in cancer patients[183].

CNS toxicity

Drug-induced neurotoxicity is a major cause for discontinuation of candidate drug development, with CNS adverse effects reported to cause 7–34% of the discontinuation of early- and late-stage projects[184]. Organoids offer a powerful model for uncovering mechanisms of toxicity that may not be detectable using more traditional approaches. Simpler systems such as 2D cell cultures may overlook unique toxicity mechanisms that depend on the intricate cellular interactions with the CNS microenvironment and, although mouse models do replicate some of the complexity and diversity of the CNS, they may fail to accurately represent human-specific responses to toxins. For example, mechanistic studies of vincristine neurotoxicity have historically focused on peripheral neurons, owing to the lack of in vitro models that resemble human brain tissues. More recently, iPS cell-derived cerebral organoids have revealed neurotoxic effects of vincristine on brain development, identifying ECM-mediated signalling and highlighting the function of matrix metalloproteinases in the process[185].

Box 2 | MCLA-158 (petosemtamab), cancer drug development using a dedicated cancer organoid biobank

The bispecific antibody MCLA-158, which targets leucine-rich repeat-containing G protein-coupled receptor 5 (LGR5) and epidermal growth factor receptor (EGFR), provides a compelling example of how organoid technology can advance drug development. The entire discovery programme was carried out using patient-derived colon cancer organoid lines. A library of more than 500 bispecific antibodies was generated, in which one arm targeted EGFR family members and the other targeted stem cell markers including LGR5. Unbiased functional screening for growth inhibition and morphological alterations in a small panel of colorectal cancer (CRC) organoid lines identified MCLA-158, which was further characterized in a broader set of healthy and CRC organoids. MCLA-158 eliminated CRC organoids but not healthy colon organoids[291].

MCLA-158 has demonstrated therapeutic properties such as KRAS-mutant CRC tumour growth inhibition, metastasis initiation blockade and suppression of tumour outgrowth in preclinical models for several epithelial cancer types. The development programme progressed from discovery to phase I within 30 months. Ongoing clinical studies include phase III registrational trials in first-line head and neck squamous cell carcinoma (HNSCC), as monotherapy[292] and in combination with pembrolizumab[293], and a phase II open-label trial in late-line metastatic CRC was started in late 2024 (ref. 294).

Table 3 | Organoid applications for drug safety assessment

Molecule tested	Key findings	Refs.
Liver		
Multiple small molecules	Established a large-scale screening platform, annotated 238 drugs and modelled genotype-specific susceptibility to bosentan-induced cholestasis	164
Acetaminophen, fialuridine, methotrexate, TAK-875	Modelled DILI phenotypes such as steatosis, fibrosis and immune responses with clinical concordance	165
Troglitazone, paracetamol, trovafloxacin, levofloxacin	Generated mature hepatic organoids and assessed hepatotoxicity using markers such as ROS and mitochondrial respiration	82
Chlorpromazine	Demonstrated bile duct toxicity mechanisms using cholangiocyte organoids	166
Multiple small molecules	Modelled phospholipidosis better than HepG2 cells, highlighting metabolite-mediated toxicity	163
Kidney		
Cisplatin	Differential apoptosis observed in response to cisplatin toxicity	18
Gentamicin, cisplatin	Organized nephron-like structures; modelled acute kidney injury with dose-dependent KIM1 expression	169
Gentamicin, cisplatin	Validated similar KIM1 expression in response to gentamicin and cisplatin	170
Cisplatin	Recapitulated kidney injury caused by cisplatin treatment	171
Doxorubicin	Demonstrated dose-dependent apoptosis with doxorubicin treatment	173
Adriamycin	Modelled adriamycin-induced toxicity in micro-organoids	174
Cisplatin	Used kidney organoids to model cisplatin-induced injury	172
Doxorubicin, aminoglycosides	Bioprinted organoids recapitulated nephrotoxicity with high reproducibility	175
HLA-A2 WT1 bispecific antibody	Modelled selective killing of WT1-expressing cells using immune-infiltrated kidney organoids	176
Gut		
31 reference drugs	Developed assay for drug-induced diarrhoea with 90% accuracy	177
Doxorubicin, gefitinib	Elucidated gastrointestinal toxicity mechanisms of anticancer drugs like doxorubicin and gefitinib	178,179
Multiple small molecules	Enabled high-throughput screening with cost-reduced intestinal organoid culture	180
Multiple small molecules	Differentiated toxic and non-toxic drugs, showing species-specific sensitivity across organoids	181
T cell bispecific antibodies	Studied off-tumour toxicities of T cell bispecific antibodies using PDOs	182
T cell bispecific antibodies	Investigated inflammation from biologics using immuno-organoids with tissue-resident T cells	183
Central nervous system		
Paclitaxel	Replicated chemotherapy-induced neurotoxicity with induced pluripotent stem cell-derived brain organoids	187
Vincristine	Studied vincristine's effects on brain development using cerebral organoids	185
Tranylcypromine	Modelled tranylcypromine-induced brain damage at varying concentrations	186
Heart		
Doxorubicin	Organoids exposed to doxorubicin revealed decreased viability, elevated LDH and BNP levels, increased apoptosis, fibrosis and mitochondrial dysfunction	89
Doxorubicin	Organoids exposed to doxorubicin revealed reduced size, impaired rhythmicity and collagen deposition, along with fibroblast activation and early signs of endothelial-to-mesenchymal transition	190
Doxorubicin	Doxorubicin exacerbated hypoxia-sensitive damage, with impaired calcium handling and fibrosis emerging in infarcted core zones	191
A panel of FDA-recalled compounds	Organoids exhibited dose-dependent ATP depletion and beat disruption more effectively than 2D monolayers	192
Compounds targeting GSK3, TGFβ and mevalonate pathway	The agents stimulated cardiomyocyte proliferation but induced prolonged contraction or decreased force	193
Thalidomide	Organoids exposed to thalidomide revealed reduced cardiac tissue formation and impaired function	194
Clomipramine	Clomipramine metabolism resulted in reduced beating, calcium flux disruption, and cell death in cardiac organoids	195
Capecitabine, ifosfamide	Liver metabolism of capecitabine and ifosfamide induced cardiotoxic responses downstream	196

BNP, brain natriuretic peptide; DILI, drug-induced liver injury; KIM1, kidney injury molecule 1, LDH, lactate dehydrogenase; PDO, patient-derived organoid; ROS, reactive oxygen species; TGFβ, transforming growth factor-β.

These findings underscore the importance of using complex 3D brain organoid systems to explore toxicity mechanisms that may involve the structural and cellular complexity of developing human brain tissue. Brain organoids can be cultured over long time periods to allow functional maturation. Mature cerebral organoids have been applied to study the impact of tranylcypromine, a drug used to treat refractory depression, on brain development, identifying dose-dependent brain damage on both neurons and astrocytes through transcriptional suppression of BHC110/LSD1-targeted genes and histone H3 lysine 4 demethylation[186]. More recently, iPS cell-derived brain organoids have been used to assess chemotherapy-induced neurotoxicity, specifically with paclitaxel, demonstrating the utility of organoids in evaluating small-molecule-induced neurotoxicity[187].

Cardiac toxicity

Despite the development and implementation of predictive preclinical assays, cardiotoxicity remains a major contributor to drug development failures and post-marketing withdrawals[188]. Human cardiac organoids, composed of stem cell-derived cardiomyocytes and supporting cell types should recapitulate essential aspects of cardiac tissue architecture, contractile function and electrophysiology to provide an alternative approach to the prediction of cardiac responses to pharmacological compounds.

Doxorubicin-induced cardiac toxicity has been modelled in cardiac organoids. Exposure to doxorubicin reduces organoid viability, increases lactate dehydrogenase (LDH) and brain natriuretic peptide (BNP) levels, and induces apoptosis, fibrosis and mitochondrial dysfunction[189]. In ECM-free models, doxorubicin treatment leads to organoid shrinkage, impaired rhythmicity, collagen deposition, fibroblast activation and early signs of endothelial-to-mesenchymal transition[190]. In a cardiac infarct organoid model mimicking post-ischaemic tissue, doxorubicin exacerbates hypoxia-sensitive damage, disrupting the calcium cycling required for coordinated contractions and promoting the emergence of fibrosis in infarcted core zones[191]. Cardiac organoid platforms have been used to screen a range of clinically relevant drugs. A study evaluating FDA-recalled compounds (astemizole, cisapride, mibefradil, pergolide, terodiline, rofecoxib, valdecoxib, bromfenac, tienilic acid and troglitazone) shows that organoids detect dose-dependent ATP depletion and beat disruption more effectively than 2D monolayers[192]. Additional HTS has also identified pro-proliferative compounds targeting GSK3, TGFβ and the mevalonate pathway[193] that stimulate cardiomyocyte proliferation but simultaneously cause prolonged contraction or decreased force, indicating functional side effects despite regenerative potential. Spatially organized cardiac organoids have been used to model developmental toxicity. Exposure to thalidomide reduces cardiac tissue formation and impairs function, with larger constructs showing more stable beating profiles, and smaller ones showing delayed relaxation and morphological variability[194]. Finally, multi-organ platforms have highlighted how metabolism in other tissues can drive cardiotoxicity. In liver–heart co-cultures, hepatic conversion of clomipramine into desmethylclomipramine through cytochrome P450 enzymes results in reduced beating, calcium flux disruption and cell death in cardiac organoids, effects not observed when liver organoids are absent, implying that hepatocytes are responsible for the observed toxicity[195]. A six-tissue platform similarly demonstrates that hepatic metabolism of capecitabine and ifosfamide induces cardiotoxic responses downstream, highlighting the role of inter-organ crosstalk in cardiac toxicity[196].

Organoid-based platforms for preclinical safety

The ability of conventional organoids to predict the safety of existing drugs is currently being explored, but a new generation of bioengineered organoid-based platforms with increased complexity is already demonstrating abilities to further improve drug safety testing (Fig. 3b). Such models incorporate vascular and immune components, enabling the study of drug effects on both tissue microenvironments and systemic responses[176,197], or built-in biosensors that detect the ATP to ADP ratio in real time to assess drug-induced nephrotoxicity[198]. Additionally, multi-organoid systems combining liver, heart, lung, vasculature, testes, colon and brain have been used in 'body-on-a-chip' platforms, enabling the screening of FDA-recalled drugs and offering insights into multi-organ toxicity and systemic interactions[192]. Similarly, micro-engineered assembloids of liver and islet organoids are applied to measure metformin activities[199,200].

More recently, bioengineered 2D organoids, such as mini-colon- and mini-intestine-on-a-chip platforms, have further expanded drug safety assessment capabilities by maintaining organoid differentiation and apical–basal polarity while facilitating luminal access through the fluid flow (Fig. 3a). In this set-up, 3D organoids are 'opened up' and with their outside or basal surface are placed onto a 2D substrate. The inside/luminal/apical surface of the organoid is freely accessible from above. This has proved particularly effective for testing stem cell-targeting cancer chemotherapies such as cytarabine and idasanutlin[201]. These advanced platforms that combine organoid biology with bioengineering approaches offer new options to tackle the translational safety challenges posed by therapeutics with increasingly complex mechanisms of action. Additionally, transwell-based organoid monolayers also enable quantitative assessment of epithelial barrier integrity using trans-epithelial electrical resistance (TEER) measurements, providing a non-invasive and dynamic readout of tissue polarization and tight junction functionality[181,202–205]. Notably, as an alternative to TEER measurement, image-based assays leveraging diffusion of fluorescent dyes across the tissue barrier have also been used to determine barrier integrity of intact intestinal organoids in vitro, either by adding a fluorescent dye to the culture medium and following its internalization upon barrier degradation[206] or by directly micro-injecting a dye inside the luminal space of the organoid and quantifying the epithelial barrier leakiness[207].

Organoid-based absorption and metabolism assays

The ability to predict drug absorption, distribution and metabolism is key to drug development, directly influencing the pharmacokinetics and therefore the predicted efficacy, safety and dose of therapeutic candidates. Understanding these parameters is essential to optimize drug formulations, mitigate adverse effects and ensure clinical success. The liver, intestine and kidney are the principal organs responsible for drug absorption, metabolism and clearance[208,209] and, as such, the main in vitro organ models of interest for the pharma industry[210].

Liver organoids for drug metabolism

The liver is the primary organ for drug metabolism, facilitating three key processes: active uptake of certain drugs to the liver, biotransformation into metabolites, and the secretion of drugs and metabolites into the bile. Current state-of-the-art in vitro models, such as cryopreserved primary hepatocytes in short-term suspension or hepatocyte co-culture systems, can be used to determine intrinsic clearance and active drug uptake. However, these models face limitations such as the inability to fully replicate hepatic metabolism and transport phenotypes (primary human hepatocytes deteriorate rapidly in culture), suboptimal cell to medium ratios that affect the measurability of metabolism and transport processes, and limited capacity to evaluate biliary secretion. Liver cell lines express suboptimal levels of drug-metabolizing enzymes and transporters, limiting their ability to predict human pharmacokinetics[211]. To address this, a TSC-derived liver organoid system has been developed that expresses a broad range of phase I drug-metabolizing enzymes (such as the cytochrome P450 enzymes CYP3A4 and CYP2D6, which modify drugs chemically), phase II enzymes (such as UDP-glucuronosyltransferases and sulfotransferases, which conjugate these metabolites to facilitate their excretion) and drug transporters (including anion transporter OATP1B3, ATP-dependent translocase ABCB1 (also known as MDR1) and MRP3). This system, integrated with a millifluidic chip, has been applied to study the metabolism and bioavailability of midazolam, as well as the excretion of coumarin and metformin in a multi-organ platform combining intestine and kidney organoids[212]. Furthermore, intrahepatic cholangiocyte organoids (ICOs) have been used to evaluate metabolite formation for paracetamol, hydroxymidazolam and 7-hydroxycoumarin glucuronide. Although the ICOs successfully model metabolite production, their CYP enzyme expression levels are lower than those observed in primary human hepatocytes, highlighting the need for further optimization to match primary tissue functionality[213].

Intestinal organoids for drug metabolism and transport

Early proof-of-concept studies have illustrated how intestinal organoid systems contribute to a better understanding and prediction of intestinal absorption and metabolism. Activities of CYP enzymes can be successfully induced in TSC-derived intestinal and liver organoids by dexamethasone, β-naphthoflavone and the constitutive androstane receptor agonist TCPOBOP, enabling drug metabolism studies[214]. Indeed, mouse-derived crypt organoids have been used to investigate intestinal drug-metabolizing enzymes (DMEs), showing the conversion of the prodrug CPT-11 into its toxic metabolite SN-38 (ref. 215). Human duodenal organoids are applied to study peptide-like drug transport, demonstrating the uptake of cefadroxil, β-lactam antibiotics and cyclic hexapeptides[216]. Moreover, a transwell-based 2D monolayer system facilitates bidirectional (apical/basal) transport studies (Fig. 3a). As described above, this 2D set-up starts from 3D organoids, which are opened up and placed on a 2D substrate. Whereas delivery of a compound to the apical side of epithelial cells in 3D organoids requires injection into the organoid lumen, the 2D transwell set-up allows easy access to both the apical and basal sides of the epithelial structure (Fig. 3a). Thus, a monolayer culture of polarized human duodenal organoids enriched in enterocytes successfully discriminated between high- and low-permeability compounds, showing functional activity of major efflux transporters, including MDR1 and BCRP[181]. Additionally, 3D duodenal and colonic organoids exhibit metabolic activity for phase I and phase II DMEs, with segment-specific differences aligning with known DME expression profiles[181]. To allow direct access to the apical surface, PS cell-derived intestinal organoids with reversed epithelial polarity have been generated, which facilitate the study of transporter and enzyme activity. These 'inside-out' organoids (Fig. 3a) show functional induction of CYP3A4 and MDR1 following exposure to rifampicin and calcitriol, and inhibition by verapamil and ivacaftor[217].

Kidney organoids for drug transport and clearance

Despite advancements in complex kidney-on-a-chip systems (reviewed in ref. 218), bioprinted kidney models[219,220] and vascularized organoid-on-a-chip platforms[197,221], practical applications of kidney organoids for the study of drug transport and clearance remain limited, with few compelling-use case examples. An organoid-derived proximal tubule epithelial cell-on-a-chip model has shown important upregulation of key transporters, including organic cation transporter 2 (OCT2) and organic anion transporter 1 (OAT1) and OAT3. This enhanced transporter expression results in improved drug uptake compared with control chips that use immortalized proximal tubule epithelial cells[222]. Similarly, a tubuloid-on-a-chip platform based on urine-derived kidney organoids replicates a functional kidney tubule barrier within a microfluidic system. Using fluorescent rhodamine 123, this model has demonstrated OCT1 and OCT2-mediated transport, highlighting the utility of this platform in the study of P-glycoprotein activity and trans-epithelial drug transport[223].

Adoption of organoids for ADME

The application of organoids in the absorption, distribution, metabolism, and excretion (ADME) space remains limited. Key challenges include the relative lack of maturity of liver and kidney organoid differentiation. Additionally, the relative simplicity of conducting small pharmacokinetic studies in animals, despite translatability caveats, often outweighs the perceived benefits of organoid-based platform predictions. Addressing ADME parameters of a drug demands a platform with enzyme activity representative of the in vivo situation, barrier functions that recapitulate the physiological functions in humans and sufficient tissue mass (millions of cells) to catalyse the biotransformation of drugs. Nevertheless, organoids are beginning to be explored to address ADME-related questions. Despite technical challenges, organoids offer promise in meeting ADME testing needs and addressing key industry hurdles in predicting drug metabolism across modalities. For small molecules, their key advantage lies in recapitulating relevant transport, barrier functions and physiological metabolic activities. Recent advances in liver organoids with functional hepatobiliary structures[79,224,225] could help to address crucial gaps in assessing drug biliary clearance, and progress in the development of kidney organoids[226–229] could be leveraged to enhance the preclinical evaluation of drug renal secretion. The ability of organoids to faithfully replicate tissue target expression is particularly appealing for assessment of large molecule binding, target evaluation and target-mediated drug disposition. A combination of fluidically linked organoid tissues and vascular organoids holds potential to tackle pharmacokinetic and tissue distribution challenges by enabling repeated sampling and precise evaluation of drug metabolism and delivery to target tissues[230–233].

Potential solutions to current challenges

As outlined above, organoid technology still faces hurdles that limit its widespread application in drug development. One hurdle is constituted by the fact that organoids, although much more complex than cell lines, still present a reductionist model and lack essential elements of real tissues. Most organoids lack blood vessels, which limits nutrient and oxygen delivery and consequently affects growth, maturation and function, especially in larger organoids. As most drugs are delivered intravenously, the absence of vascularization in organoid models poses challenges for accurate drug evaluation. Organoid cultures typically do not include immune cells, and therefore the analysis of immune responses, inflammation or immunotherapy interactions is challenging. Other complex tissue components, such as ECM, stromal cells and neurons, are also absent from organoids, impacting their functionality and disease modelling capability. Organoids do not model systemic interactions such as hormonal signalling or multi-organ crosstalk. Primary tissues experience biomechanical forces, such as shear stress, stretching and pressure, that are hard to recapitulate in organoid cultures. Another major hurdle is the variability of organoid cultures, which can lead to difficulties in achieving reproducibility: variations in protocols between labs, such as differences in cell source, media composition and growth condition, can lead to inconsistencies in organoid development; the genetic diversity of cells sourced from different patient donors can lead to variability between organoid batches; achieving uniform and controlled differentiation or functionality is still challenging, and leads to heterogeneity within and between organoid batches; and data analysis often requires sophisticated and customized analytical tools, also adding complexity to the research process and conclusion. These limitations in turn complicate the use of organoids as standardized models for HTS and large-scale drug testing. Along the same lines, scalability remains a challenge for drug development. Overcoming these challenges will require advances in culture protocols, the integration of microfluidic technologies and improvements in model complexity, which together will enable organoids to fully realize their potential as a reliable tool in personalized medicine and drug discovery.

A conventional procedure for organoid-based drug screening involves embedding organoids in a matrix and seeding these into multiwell plates, with culture medium added on top. Automated robotics and liquid-handling systems can streamline this process (Fig. 4a). This method has been used in drug screening for PDTOs derived from breast[144] and colorectal cancers (CRCs)[140]. However, the approach lacks precise control over the quantity and spatial distribution of the seeding organoids, limiting the reproducibility of HTS assays.

Bioengineered microfluidic devices enable the rapid generation of numerous replicate organoids embedded in a small volume of ECM such as Matrigel — a solubilized basement membrane extract from the Engelbreth–Holm–Swarm mouse sarcoma rich in ECM proteins (laminin 111, collagen IV and fibronectin) that supports polarity and survival. These devices seed organoids in multiwell plates, microwells/microarrays or micropillar chips (Fig. 4a–d) allowing precise control of organoid shape, size and cell numbers, which greatly enhances homogeneity and reproducibility (Fig. 4b). Such systems are well suited for HTS, for which assays need to be simple and robust. Preferably, the generated organoids are positioned at uniform horizontal and vertical planes, facilitating imaging-based phenotypic analysis. For example, micro-engineered hydrogel films embedded in multiwell plates can simultaneously generate thousands of uniform organoids trapped in the microcavity arrays on the same focal plane, enabling real-time automated imaging for thousands of individual CRC organoids treated with anticancer drugs[234]. High-throughput printing of Matrigel-embedded organoid droplets using microfluidics ensures inter-organoid homogeneity while retaining tissue heterogeneity, offering robust drug response predictions aligned with patient outcomes[235,236]. The small droplet format — with its large surface-to-volume ratio — also enables efficient virus infection and T cell penetration, and has been applied to drug testing for SARS-CoV-2 and rapid chimeric antigen receptor (CAR)-T potency assays[235]. Microwell arrays (Fig. 4c) have been used to analyse lung cancer organoids, predicting drug response within a week[237], whereas 3D-cultured neural stem cells grown in 384-micropillar plates (Fig. 4d) have enabled neurotoxicity screening across large chemical libraries[238].

These 3D high-throughput platforms typically use biocompatible coating matrices, such as polydimethylsiloxane[239], polymethyl methacrylate[240], hydrogel[241] or Matrigel[242], to maintain organoid 3D architecture and support cell growth. Matrix-free suspension cultures using ultra-low attachment plates with rotor devices can be applied for certain organoid types, such as brain organoids[121] and liver assembloids[243]. These suspension cultures can also be integrated into microfluidic systems to generate uniform organoids of controlled sizes for HTS assays.

Approaches for automation

Early developments in automation have further improved the integration of organoids into microfluidics-based HTS platforms (Fig. 4e). One example, the Mimetas organoplate, enables automated medium changes, precise control over environmental conditions and even drug screening with minimal human intervention. The platform has been used to test liver organoids for hepatotoxicity[244] and lung organoids for modelling respiratory disease[245], demonstrating the potential of automation in both drug screening and disease modelling. Another early organoid-on-a-chip approach has been introduced by Emulate[246]. The orbital shaker platform provides automated organoid culture and maintains constant, gentle agitation, which is crucial for optimal growth and uniformity in 3D organoid cultures[247]. Such systems can be integrated with multiwell plates for upscaling. Another notable example of automation in organoid-based research involves culture and screening of PDTOs derived from colorectal, pancreatic and other cancers using a high-throughput, automated platform that incorporates robotic liquid handling for automated medium exchange, drug addition and compound screening, enabling rapid testing of large drug libraries[248,249].

To extract information from the tissue complexity offered by organoids, image-based technologies and high-content screening systems have been integrated into organoid automation to provide robust data acquisition and analysis. Fluorescence microscopy and automated imaging platforms such as CellInsight CX7 enable the monitoring of organoid growth, morphological changes and drug responses in real time. These systems use computer vision and image analysis algorithms to quantify parameters such as cell viability, structural integrity, gene expression and drug efficacy[250,251]. For instance, in AD research, brain organoids have been screened for the effects of drug candidates targeting amyloid plaques or tau tangles using automated microscopy systems[252].

As for the readouts of automatic HTS assays, the simplest and most reliable approaches involve quantification of fluorescence or absorbance signals (Fig. 4f), such as the widely used cell viability assays based on CellTiter-Glo[144] or MTT[253]. Data generated from plate readers can be digitally quantified and directly processed using standardized statistical methods. These principles can be extended for the design of disease-specific readouts. For example, the expression or activity of key disease proteins (for example, A1AT production in AATD) can be linked to catalysed chemical reactions or ELISA assays[56]. Disease-associated phenotypes (for example, lipid accumulation in fatty liver disease) can be visualized using fluorescent indicators and analysed through phenotypic imaging[90]. Some genetic disorders can be assayed using functional assays (for example, swelling assay mediated by CFTR function[42]). Infectious disease-related drug screening may rely on fluorescence-labelled pathogens or detection of pathogen-associated RNA or proteins[235].

In addition, advanced histotechnology, a highly standardized method used in diagnostic and toxicological pathology with

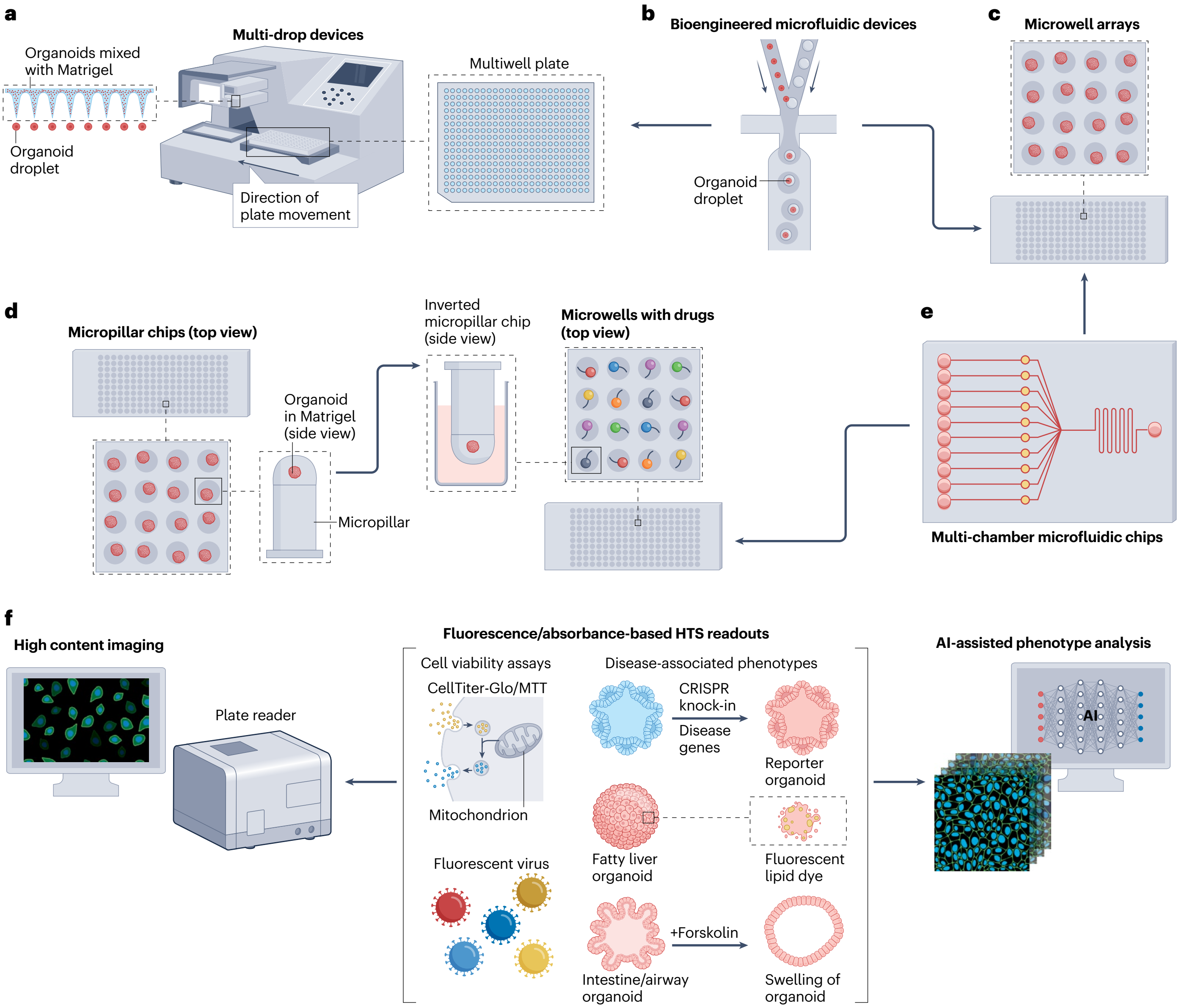

Fig. 4 | Strategies for automated high-throughput drug screening. **a**, A conventional method that uses multi-drop devices to automate the seeding of organoid–Matrigel mixtures into multiwell plates. **b**, Bioengineered microfluidic devices facilitate the rapid generation of numerous replicate organoid droplets, with precise control over their shape, size and cell numbers. **c**,**d**, These organoid droplets can subsequently be introduced into multiwell plates (part **a**), microwell arrays (part **c**) or micropillar chips (part **d**) for further drug evaluation. Different colours indicate different drugs in the microwells. **e**, A multi-chamber microfluidic chip can be used for high-throughput screening (HTS) by simply seeding organoid cells directly into the chambers, or it can be seamlessly integrated into existing HTS set-ups (that is, microwell or micropillar chips) to facilitate drug delivery. **f**, Automated imaging and data processing based on fluorescence or absorbance readouts, with the assistance of artificial intelligence (AI)-driven algorithms. Examples of the HTS readouts include cell viability assays, such as CellTiter-Glo or MTT, fluorescent pathogens used for drug screening in infectious diseases, and assessments based on disease-associated phenotypes, such as the expression of disease markers, the activity of disease enzymes, fluorescent dyes to monitor metabolite dynamics in metabolic disease and morphological changes related to the disease.

morphological end points, is being progressively adopted to increase throughput and standardization in organoid research. Organoids can be fixed in formalin and embedded in paraffin (FFPE) or are fresh frozen embedded in optimal cutting temperature compound or gelatine[254,255]. Tissue microarrays[256], which allow a full 96-well plate of organoids to be embedded in an FFPE block, provide a great tool for artificial intelligence (AI)-supported quantitative image analysis, adapted from the emerging field of digital pathology[257]. This enables the application of molecular pathology methods — including immunohistochemistry, multiplex immunofluorescence, spatial transcriptomics and electron microscopy as well as matrix-assisted laser desorption/ionization (MALDI) imaging – to organoid systems[182,254].

Regulatory challenges and opportunities

Although organoids offer considerable advantages, regulatory acceptance remains a major hurdle before they can be seamlessly integrated into the drug development pipeline. Addressing these challenges could strengthen drug safety assessment, enable the development of more effective therapies and reduce reliance on animal testing.

Adoption of organoids in drug safety testing

The main challenges for organoid adoption in drug safety testing remain model qualification, reproducibility, clinical translation, cost–benefit optimization and regulatory acceptance. To support first-in-human clinical trials, preclinical testing typically combines in vivo and in vitro models to maximize clinical predictivity while minimizing animal use. Organoids can supplement traditional 2D cell culture assays when functional and physiologically relevant end points are needed, particularly to address mechanistic questions[29,258], but in vitro tools with a low level of complexity, higher throughput and lower cost are currently favoured for regulatory packages. Demonstrating clear advantages over conventional methods, such as superior predictive power in specific contexts of use, is essential for broader adoption. In addition, expansion of qualification efforts across diverse contexts of use and therapeutic modalities will be crucial to increase confidence in organoid predictive value over simpler, conventional in vitro methods. Importantly, although organoids show promise as translational models for prediction of toxicity, most studies presented here represent preclinical proof-of-concept work, and validation against in vivo and patient data is still limited. Additional forward and reverse translation studies will be required to establish organoids as translational standards.

The application of organoids in preclinical drug development typically falls into two categories: internal decision-making (for example, screening and mechanistic studies) and incorporation into regulatory filings[29]. Pharmaceutical companies have used organoids for exploratory and mechanistic studies in selected contexts of use[29,258,259], but these studies are rarely pivotal in determining safety for clinical trials[259].

Organoid complexity also presents practical barriers. Their construction and maintenance demand considerable time, resources and technical expertise, often requiring advanced imaging and data analysis pipelines. For organizations with limited resources or sporadic use cases, sustaining this expertise in-house can be impractical. Outsourcing to specialized contract research organizations with a critical mass of organoid-based projects sustaining a solid business model is a viable alternative to internal efforts. In addition, efforts to standardize and automate protocols and simplify work-flows will be crucial to unlocking the broader utility of organoids and ensuring their integration into the drug development pipeline.

Regulatory framework for organoids applied in drug development

The goal of non-clinical studies before first-in-man clinical studies is to assess the pharmacological activity of the product and to ensure the safety of healthy volunteers or patients in phase I clinical studies. For human drug development, FDA regulations do not dictate the type or design of studies used (whether in vivo, in vitro or in silico) to support that a drug considered for clinical investigation is reasonably likely to be safe in humans[260]. Although animal toxicity studies have proved essential to identify potential human risks, finding ways to reduce the use of animals and develop effective alternatives is an important effort[260,261]. There is a huge demand for more human-predictive translational platforms and technologies, such as organoids, in the regulatory context defined as new approach methodologies (NAMs)[29,262]. US Congress passed the FDA Modernization Act 2.0 in 2022 (ref. 30), followed by Act 3.0 (ref. 31), clarifying the perceived requirement of animal testing in drug development. This requires the FDA to establish a pathway by which entities may apply to have non-clinical testing methods approved for use in a particular context of use. For this, the FDA launched the 'Alternative Methods Program'[263], first, to expand processes to qualify alternative methods for regulatory use; second, to provide clear guidelines to external stakeholders developing alternative methods; and third, to fill information gaps with applied research to advance new policy and guidance development[260]. More recently, the FDA has announced their plan to replace animal testing in the development of monoclonal antibody therapies and other drugs with a range of approaches in a laboratory setting, including AI-based computational models of toxicity and cell lines and organoid toxicity testing[32].

However, successful implementation of organoids into preclinical testing pipelines requires their proper characterization, validation and qualification to demonstrate to end-users, as well as health authorities, that they are fit for a given purpose or context of use[259,264–266] (Box 3). Principles of validation are well described by the Organization for Economic Cooperation and Development (OECD), which provides useful guidance for how organoids might be evaluated for individual contexts of use[267–270]. Such an assessment is standard for methods used in the regulatory environment[262,265]. Efforts to achieve regulatory acceptance of NAMs, including organoids, are currently being co-developed by cross-industry consortia and health authorities, focusing on defining standards and training stakeholders[29,259,261]. Ideally, diverse teams of scientists, including bioengineers, biologists, toxicologists and pathologists collaboratively develop organoid models and collectively assess their context of use[254].

Regulatory outlook

Currently, only a few cell-based test methods are described by the OECD test guidelines. Most toxicity tests still fall within the category of non-guideline methods[271]. Recently, both the FDA and EMA have proposed road maps to advance the regulatory acceptance of NAMs

Box 3 | Regulatory guidelines

At the Organization for Economic Cooperation and Development (OECD) level, there are three relevant guidance documents for in vitro method development and use: the *Guidance Document on Good In Vitro Method Practices (GIVIMP)*[268], the *Guidance Document for Describing Non-Guideline In Vitro Test Methods* (GD211)[269] and the *Guidance Document on the Validation and International Acceptance of New or Updated Test Methods for Hazard Assessments* (GD34)[267]. GD34 and GD211 support the thorough description, characterization and validation of organoids to enhance their credibility. Whereas characterization can include an assessment of the biological relevance and scientific validity of the organoids, validation refers to the process by which the reliability and relevance of a particular assay is established for a defined context of use[267,270].

for drug development, for example, the EMA's voluntary submission of data procedure, also known as the safe harbour approach, aims to allow the generation, compilation and review of data to help define a context of use for a NAM[272,273]. In addition, the EMA evaluates the readiness and limitations for regulatory acceptance of a given NAM within a specific context of use. This process aims to build confidence in NAMs and may support the drafting of qualification criteria based on defined contexts of use, which in turn would guide model developers on how to qualify novel models for regulatory applications[274]. The FDA's road map focuses on replacing animal testing for monoclonal antibodies and other drugs over a period of several years[32,272].

Therefore, progressively incorporating organoids as a NAM in regulatory submissions to replace, reduce and refine animal models offers potential. However, this should not compromise the established standards required for the assessment of medicines. It is essential to maintain a careful equilibrium – embracing these innovative methodologies, including organoids, while upholding the rigorous standards of the European Medicine Regulatory Network.

Conclusions

The use of organoids in drug development offers a transformative leap towards more accurate and patient-relevant models in preclinical research. As described, organoids represent a platform for recapitulation of the complexity of human tissues, providing a more physiologically relevant environment than traditional 2D cell culture and animal models. However, although the field has made important strides, challenges remain. The scalability and reproducibility of organoid cultures, particularly in the context of HTS, need further refinement. These will require continued development and validation of standardized, automated organoid platforms and clear regulatory frameworks that facilitate industry adoption. Additionally, enhancing the predictive power of organoids for human therapeutic responses – especially in complex diseases such as cancer, neurodegenerative disorders and chronic inflammatory and metabolic diseases – will also require innovations in vascularization, immune system integration and multi-organ modelling, together with cutting-edge analytical technologies, such as single-cell omics, high-speed image acquisition and AI-supported data analysis. As the technology matures, we expect organoid-based platforms to become an integral part of the drug discovery pipeline, bridging the gap between basic science and clinical outcomes. Organoids may revolutionize the way we understand and treat disease, and their future in drug development holds great potential.

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Author contributions

D.W., R.V., and N.S.-R. researched data for article. All authors provided substantial contribution to discussion of content, wrote and reviewed and edited the article before submission.

Competing interests

H.C. was the head of Pharma Research and Early Development (pRED) at Roche, Basel, and holds several patents related to organoid technology. His full disclosure can be found at: www.uu.nl/staff/JCClevers/Additional functions. R.V. and N.S.-R. are employees and stockholders of F. Hoffmann-La Roche Ldt.

Additional information

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